

Government Document Anomaly Detection System

An AI-assisted system for detecting potential financial anomalies in publicly available government documents such as budgets, procurement records, expenditure reports, and audit reports.

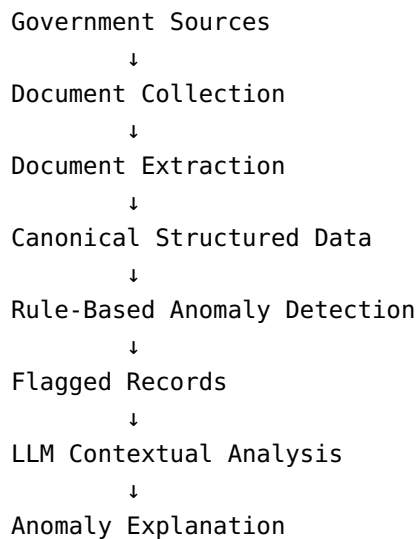
Project Objective

The goal of this project is to automatically analyze government documents and identify potentially unusual financial patterns, such as:

- Low utilization of allocated budgets
- Unusual procurement prices
- Concentration of spending near the end of a financial year
- Inconsistencies between quantity, unit price, and total value

The system is designed to identify potential anomalies, not to determine wrongdoing.

System Architecture



Development Phases

Phase 0 — MVP

- Select one government department
- Identify verified government sources

- Collect 5–10 documents
- Manually extract important fields
- Create the canonical dataset
- Implement rule-based anomaly detection

Phase 1 — Automated Extraction

- Automate document extraction
- Add LLM-based structured extraction
- Build market-price benchmark data
- Add statistical outlier detection

Phase 2 — LLM Analysis Pipeline

- LLM extraction
- Rule-based flagging
- Contextual adjudication
- Explanation generation
- Entity resolution

Phase 3 — Expansion

- Add additional departments
- Compare departments and years
- Add human-review workflow
- Build a user-facing dashboard

Initial Scope

The initial implementation will focus on one government department and a small number of document types before expanding to additional sources.

Canonical Data Schema

The extracted information will eventually be normalized into fields such as:

- department
- scheme_name
- fiscal_year
- budget_allocated
- funds_released
- expenditure_actual
- item_description
- quantity
- unit_price
- total_value
- vendor_name

- award_date
- completion_date
- source_page

Anomaly Detection

Initial deterministic rules will include:

1. Low budget utilization
2. Procurement price deviation
3. Spending concentration near the end of the financial year
4. Quantity × unit price inconsistency

Technology Stack

- Python
- Pandas
- PDF extraction tools
- OCR
- LLM APIs
- Rule-based analysis
- Statistical analysis
- Git/GitHub

Project Status

🚧 Currently in development.

Completed

- [x] Initial project planning
- [] Government source verification
- [] Dataset creation
- [] Manual extraction
- [] Rule-based anomaly detection
- [] Automated extraction
- [] LLM contextual analysis
- [] Dashboard

Important Principle

The system is intended to flag potential anomalies for further investigation. A detected anomaly does not imply fraud, corruption, or wrongdoing.

Project Structure

```
government-document-anomaly-detection/  
|  
├─ data/  
├─ docs/  
├─ src/  
├─ tests/  
├─ notebooks/  
├─ app/  
├─ README.md  
├─ requirements.txt  
└─ .gitignore
```